

Interpret Probability in Data-Rich and Realistic Contexts

Step 11

 Upper Elementary
 (Grades 4–5)

 Today I am learning to interpret probability
 in data-rich and realistic contexts.

1. Weather Data

The table shows the weather recorded in a city over 40 days.

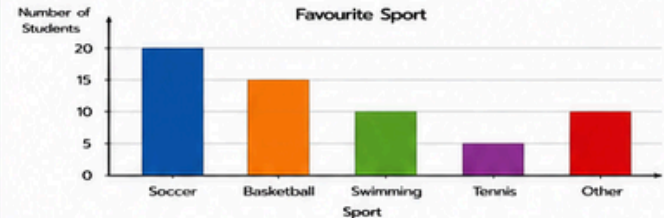
Weather Type	Sunny	Cloudy	Rainy	Windy
Number of Days	18	10	8	4

- a) What is the experimental probability of each type of weather?
 Write as a fraction and as a percentage.
- Sunny: _____ (____%) Cloudy: _____ (____%)
 Rainy: _____ (____%) Windy: _____ (____%)
- b) Based on this data, which type of weather is most likely tomorrow?
 Which is least likely?
- Most likely: _____ Least likely: _____
- c) The weather forecaster says there is a 20% chance of rain tomorrow.
 Based on the data, what do you think of this forecast?
- _____



2. Favourite Sports Survey

A class of 60 students were asked about their favourite sport.



- a) What is the probability that a randomly chosen student's favourite sport is:
- Soccer _____ Basketball _____ Swimming _____
 Tennis _____ Other _____
- b) If 3 students are chosen at random, estimate how many would you expect
 to have soccer as their favourite sport. _____
- c) If another student is added who loves tennis, how would this change
 the probabilities? _____

3. Bag of Counters – Data and Probability

A bag contains counters of different colours.



Colour	Red	Blue	Green	Yellow
Number in Bag	12	8	6	4
Experimental Probability				
Fraction				
Percentage				

- a) Complete the table.
- b) If a counter is picked at random, which colour is twice as likely as yellow?

- c) A counter is picked 30 times. Based on the experimental probabilities,
 how many of each colour would you expect to be picked?
- Red: _____ Blue: _____ Green: _____ Yellow: _____

4. Tickets in a Raffle

A school raffle has 500 tickets.

The table shows how many tickets were sold.



Ticket Type	Number Sold
Blue	150
Red	200
Green	100
Yellow	50

- a) What is the probability of each ticket type
 winning the raffle? Write as a fraction
 and as a percentage.
- Blue: _____ (____%) Red: _____ (____%)
 Green: _____ (____%) Yellow: _____ (____%)
- b) If you buy 1 blue ticket, how does your chance
 of winning compare to someone who buys 1 yellow ticket?

- c) If you buy 3 tickets – one of each colour – how does this change your
 chance of winning compared to buying just 1 ticket?

5. Health Data

A health survey of 1 000 people found the following:

- 340 people exercise regularly
- 280 people eat at least 5 serves of vegetables each day
- 160 people do both
- The rest do neither

a) Complete the two-way table.

	Exercise Regularly	Do Not Exercise Regularly	Total
Eat 5+ Serves of Vegetables	_____	_____	_____
Do Not Eat 5+ Serves	_____	_____	_____
Total	_____	_____	1 000

- b) What is the probability that a
 randomly chosen person:
- i. exercises regularly and eats
 5+ serves of vegetables?

- ii. exercises but does not eat
 5+ serves of vegetables?

- iii. does neither?



6. Bus Arrivals

The table shows how often a bus arrives within a certain time after it is due.

Time After Due	On Time (0–2 min)	3–5 min	6–10 min	More than 10 min
Number of Arrivals (out of 200)	80	60	40	20

- a) What is the probability that the bus will arrive:
- i. on time? _____ ii. within 10 minutes? _____
 iii. more than 10 minutes late? _____
- b) If you wait for the bus 4 times next week, how many times would you expect
 it to arrive more than 10 minutes late? _____
- c) Is the statement "The bus is usually on time" supported by the data? Explain.



7. Comparing Two Events

Use the data to compare the events.

Event	Experimental Probability
A: Rolling a 6 on a number cube	$\frac{1}{6}$
B: Drawing a heart from a deck of cards	$\frac{1}{4}$
C: Spinning red on this spinner	$\frac{3}{8}$



- a) Order the events from least likely to most likely.
 _____ < _____ < _____
- b) Which two events have the biggest difference
 in probability?

- c) In what real-life situation might event C happen?

8. Think and Explain

- a) Why is it important to use real data (like surveys or records)
 to understand probability?

- b) Give an example from your own life where you used chance
 to make a decision.

- c) How can both data and probability help us make better choices?



I can...

- ☐ calculate experimental probabilities from data.
- ☐ compare probabilities in real-life contexts.
- ☐ use data to make predictions.
- ☐ explain and justify my reasoning.
- ☐ use probability to judge risk and uncertainty.

Chance Words



Great job!

 You are using probability
 to understand the world
 and make smarter decisions!